Chapter-2

Friction, Work and Gravitation

Textbook: Fundamentals of Physics by Halliday, Resnick & Walker [extended 10th edition]

Topics to be covered:

6-1: Friction, 7-2: Work and Kinetic energy, 7-3: Work done by Gravitational force, 7-4: Work done by spring force, 13-1: Newton's Law of Gravitation, 13-2: Gravitation and principle of superposition, 13-3: Gravitation near earth's surface, 13-4: Gravitation inside earth.

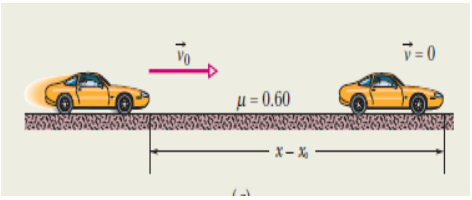
Theory

1. Define: (a) Friction, (b) Static Friction, (c) Kinetic friction and (d) spring force
2. What are the properties of friction?
3. Explain work. State and prove the work-kinetic energy theorem.
4. Derive the expression for work done by spring force.
5. State Newton's Law of Gravitation.
6. What is the acceleration due to gravity, g. What are the reasons for the variation of g over the earth surface?
7. Find the expression for the gravitational acceleration inside a uniform Earth at a distance r from the center [$g_r = \frac{GM}{R^3}\, r$].
8. Define escape velocity. State and explain the Kepler's law.

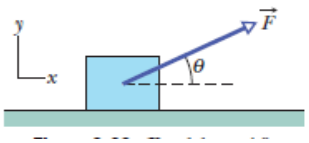
Friction (Chapter-6)

1. [1] The floor of a railroad flatcar is loaded with loose crates having a coefficient of static friction of 0.25 with the floor. If the train is initially moving at a speed of 48 km/h, in how short a distance can the train be stopped at constant acceleration without causing the crates to slide over the floor?

2. [Sample Problem-6.02] How far does the car take to slide to a stop on a horizontal road (Figure) if the coefficient of kinetic friction is μ_k = 0.60? Let's neglect any effect of the air on the car, assume that the wheels lock up and the tires slide, and extend an *x* axis in the car's direction of motion.



3. [10] Figure shows an initially stationary block of mass *m* on a floor. A force of magnitude 0.500*mg* is then applied at upward angle θ= 20°. What is the magnitude of the acceleration of the block across the floor if the friction coefficients are μ_s= 0.600 and μ_k=0.500.

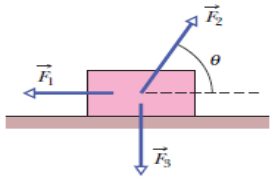


4. [11] A 68 kg crate is dragged across a floor by pulling on a rope attached to the crate and inclined 15° above the horizontal. (a) If the coefficient of static friction is 0.50, what minimum force magnitude is required from the rope to start the crate moving? (b) If μ_k = 0.35, what is the magnitude of the initial acceleration of the crate?

Work (Chapter-7)

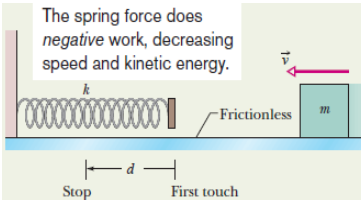
5. [13] A luge and its rider, with a total mass of 85 kg, emerge from a downhill track onto a horizontal straight track with an initial speed of 37 m/s. If a force slows them to a stop at a constant rate of 2.0 m/s^2, (a) what magnitude F is required for the force, (b) what distance d do they travel while slowing, and (c) what work W is done on them by the force?

6. [15] Figure shows three forces applied to a trunk that moves leftward by 3.00 m over a frictionless floor. The force magnitudes are F_1= 5.00 N, F_2= 9.00 N, and F_3= 3.00 N, and the indicated angle is Θ= 60.0°. During the displacement, (a) what is the net work done on the trunk by the three forces and (b) does the kinetic energy of the trunk increase or decrease?



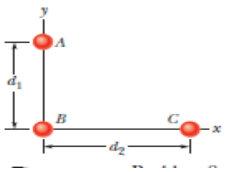
7. [17] A helicopter lifts a 72 kg astronaut 15 m vertically from the ocean by means of a cable. The acceleration of the astronaut is $g/10$. How much work is done on the astronaut by (a) the force from the helicopter and (b) the gravitational force on her? (c) Just before she reaches the helicopter, what is her kinetic energy?

8. [Sample problem] In the Figure, a cumin canister of mass m=0.40 kg slides across a horizontal frictionless counter with speed v= 0.50m/s. It then runs into and compresses a spring of spring constant k = 750 N/m. When the canister is momentarily stopped by the spring, by what distance d is the spring compressed?



Gravitation (Chapter-13)

9. [8] In Figure, three 5.00kg spheres are located at distances d_1=0.300m and d_2 = 0.400m. What are the (a) magnitude and (b) direction (relative to the positive direction of the x axis) of the net gravitational force on sphere B due to spheres A and C ?



10. [26] A uniform solid sphere of radius R produces a gravitational acceleration of g_R on its surface. At what distance from the sphere's center is there a point inside the sphere where the gravitational acceleration is $g_r = \frac{g_R}{3}$?

11. [28] Assume a planet is a uniform sphere of radius R that (somehow) has a narrow radial tunnel through its center. Also assume we can position an apple anywhere along the tunnel or outside the sphere. Let F_R be the magnitude of the gravitational force on the apple when it is located at the planet's surface. How far from the surface is there a point where the magnitude is $\frac{1}{2}F_R$ if we move the apple into the tunnel?

